

# CHAPTER 3 km reanalysis — GRIB2 archive contents

**90 variables, 246 GRIB messages per hourly file.** Taken from the converter's own registry (`wrf_era5_comparison.WRF_TO_ECMWF_PARAMID`), which is what the pipeline and its sanity check use, so this document describes what is actually written. Document updated on 2026-09-18, when the WPS static file `geo_em.d02` was recovered from LRZ after having been recorded as lost. It carries the FRACTIONAL land cover, soil texture and lake data that the hourly output had reduced to a single dominant category per cell, and it added fifteen fields: `cvl`, `cvh`, `tvL`, `tvh`, `slt`, `cl`, `dl` and the eight ERA5 soil layers. Nothing was removed. Every field here is one we can defend row by row. Where the WRF quantity is not exactly the ERA5 one, it is said so below rather than hidden; where it could not be made to correspond at all, the field was dropped instead of published under a name that would mislead (see §6). This document says what is in the files. Two companions say the rest: `MISSING_VARIABLES.md` says what is not and why, and `CHAPTER_KNOWN_PROPERTIES.md` states the **known properties and biases** of what is — the places where the archive is correct and surprising, each with the measured number. Read the third one before comparing this archive against ERA5 or against observations.

## 1. What the files are

|                    |                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Model              | <b>WRF 4.1.1, domain d02, 49 vertical levels, model top 50 hPa</b>                                                          |
| Horizontal grid    | Mercator, <b>1641 × 1353</b> points, <b>3 km</b> spacing (2 220 273 points/field)                                           |
| Domain corners     | 23.472 N, 19.972 W → 60.001 N, 42.256 E; standard parallel 44.671 N                                                         |
| Earth shape        | <b>Sphere of radius 6 370 000 m</b> ( <code>shapeOfTheEarth=1</code> ) — WRF's own sphere, declared explicitly              |
| GRIB edition       | <b>GRIB2</b> , grid definition template <b>3.10</b> (Mercator), originating centre <code>ecmf</code>                        |
| Packing            | <code>grid_ccsds</code> , <b>24 bits per value</b> ; masked fields carry a proper GRIB bitmap                               |
| Parameter identity | <b>ECMWF paramId / shortName</b> throughout, so the archive is directly comparable with ERA5                                |
| File               | one file per hour, <b>~747 MB</b> (~17.9 GB/day, ~6.6 TB/year) — measured, not estimated                                    |
| Naming             | <code>ailam-an-cima-3km-<code>{year}</code>-<code>{year}</code>-1h-v1-<code>{YYYYMMDD}</code>-<code>{HH}</code>.grib</code> |
| Period             | 2019, 2024 and 2025-01-01 → 2025-06-30, hourly (2020-2023 to follow)                                                        |

## 2. Vertical levels

**13 pressure levels**, the same for every 3-D variable:

**1000, 925, 850, 700, 600, 500, 400, 300, 250, 200, 150, 100, 50 hPa**

They are the ERA5 set truncated at the model top: the 8 ERA5 levels above 50 hPa are above WRF's `P_TOP` and cannot be produced. Single-level fields sit on surface, `heightAboveGround` (2, 10, 100 and 200 m), `meanSea`, `entireAtmosphere` (column integrals) or, for the radiation budget at the top of the atmosphere, `nominalTop`.

### 3. Time convention — read this before indexing the archive

- **Instantaneous fields** use product definition template **0**: dataDate/dataTime are the valid time.
- **Accumulated fields** (tp, tirf, sf, sro, ssro, ro, and the whole radiation budget) use template **8**, as their ECMWF definitions require. They are accumulated **\*\*from 00 UTC of the same day\*\***: reference time 00 UTC, step 0–H. They are exactly zero in the 00 UTC file.
- **10fg** is the one exception: it is a **maximum over the preceding hour** (reference time H–1, step 0–1), and it is the maximum of the *resolved* 10 m wind, not a gust parameterisation.
- A file therefore mixes reference times on purpose. **\*\*Index on validity (validityDate/validityTime), never on dataTime.\*\***
- Accumulated messages carry generatingProcessIdentifier = 128 as the marker of the 00 UTC reference.

### 4. Declared approximations

Four fields are not exactly their ERA5 namesake, and are published with that said:

| field      | what ERA5 means                                                                            | what we write                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| slor       | slope of the <b>sub-grid</b> orography, a parameter of the form-drag scheme                | the slope of the <b>resolved</b> 3 km terrain, $ \text{grad } z $ . Note eccodes will show you the parameter's own name, <i>Slope of sub-gridscale orography</i> , so nothing in the message itself says this. sdor beside it <b>is</b> the ERA5 quantity, only evaluated on a 3 km cell instead of a 31 km one: the measured ratio of medians is 0.244, which is $(3/31)^{0.605}$ — a resolution rescaling, not a different definition |
| 10fg       | gust from a parameterisation (turbulent + convective)                                      | the hourly maximum of the <b>resolved</b> 10 m wind                                                                                                                                                                                                                                                                                                                                                                                     |
| IEWS, inss | the model's own stress components                                                          | the magnitude $\rho u^2$ , projected on the 10 m wind direction                                                                                                                                                                                                                                                                                                                                                                         |
| fsr        | an <b>effective</b> roughness, carrying the orographic drag the IFS parameterises at 31 km | WRF's <b>local</b> ZNT: the land-use roughness with a snow override. Over land it runs at <b>0.32–0.48 of ERA5's</b> in the median; over sea the two agree, both being Charnock. zust beside it is the model's own friction velocity and runs <b>1.6–2.1× ERA5's over land</b> (1.1× over sea) — the other half of the same story, because at 3 km the terrain-driven wind is resolved rather than smoothed                             |

| field | what ERA5 means                                                       | what we write                                                                                                                                                        |
|-------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lsm   | a <b>permanent</b> land-sea mask; sea ice is carried separately in ci | WRF's LANDMASK, which reclassifies sea-ice points as land, so the mask <b>grows in winter</b> . lsm & ~(ci > 0) recovers the ERA5 sense exactly, and ci is published |

lsm is now the **only** field that follows the model hour by hour (with al, which moves with it). The soil columns used to as well, and no longer do: under new sea ice the scheme grows a column that is not soil — swvl reached exactly 1.000 against its own soil type's porosity of 0.435 — so since 2026-09-25 they are masked on the **permanent** land of the static file, as ERA5 does. Nothing is lost that cannot be recovered: ci is published. Three more deserve a note rather than a warning:

- **skt** is not a model output here (TSK was not written): it is inverted from the upward longwave,  $LWUPB = \epsilon \sigma T^4 + (1-\epsilon) LWDN$ . The emissivity is not assumed — it was **read out of the run itself**. The radiation scheme computes a clear-sky diagnostic with the same emissivity and the same skin temperature, changing only the downward flux, so the two equations together eliminate the unknown temperature and leave  $\epsilon = 1 - (LWUPB - LWUPBC)/(LWDN - LWDNC)$ . That is an identity, not a fit. Measured this way the emissivity came out identical in every cell of a given land-use category (spread 0.0000 to four decimals), and the control is exact: over open water, where the model's skin temperature IS the sea surface temperature, it returns 0.98000.

**The accuracy is seasonal, and both numbers matter.** The reconstruction reproduces the model's own emissivity to 1e-4 on **99.98 % of land points in July and 93.05 % in January**, for a land skt error of **0.0012 K RMSE in July against 0.115-0.133 K in January and February** (p99 0.85 K, worst point 2.8 K). All of that spread is the snow rule: snow raises the emissivity to a flat 0.98 in every category, applied from 1 % snow cover upwards, and the model's own transition completes a little later than that. Over open water, where the answer is known independently, it recovers the sea surface temperature to **0.0005 K RMSE** on about 910 000 points at every timestep of every season. Note this derivation error is thirty times smaller than the physical cold bias of the land skin itself — which is the separate and much larger fact: **skt over land runs about 1.6 K below ERA5, reaching 3.2 K at midday**, while over sea it is exact (+0.01 K). That is the run, not the derivation; the sea number is the control that proves it. Reconciling it against the extra shortwave and the missing cloud above would need the surface energy budget, and HFX, LH and the ground heat flux are the three fields this archive does not have.

- **al** is a two-season table value, not a field. No monthly satellite albedo was used (usemonalb = .false.), so the snow-free background comes from the land-use table's WINTER and SUMMER columns by dominant category: it takes only **8 to 11 distinct values** over the whole domain, and it changes at **\*\*1 171 937 land points** — 89.9 % of the land mask — at the seasonal boundary\*\*, which is April and November, not a physical transition. Off land it is 0.08 on water and 0.65 on sea ice. Do not read it as an observed or evolving albedo, and do not difference it across the boundary expecting a signal. fal, beside it, is the actual ratio the simulation used and does evolve.
- **The surface shortwave carries a systematic high bias.** RRTMG ran with an ozone climatology and **no aerosol at all** (aer\_opt = 0), while ERA5 carries the Tegen aerosol climatology; over this domain an optical depth of 0.1 to 0.2 accounts for the size. Measured against ERA5: **+8.5 % under clear sky (ssrdc)** and **+13.6 % all-sky (ssrd)**. The thermal fields are untouched by it and agree to within 1.1 %, which is the control that makes the attribution credible. Total cloud cover also runs **22 % below** ERA5 and high cloud **39 % below**, which pushes surface solar the same way. Both are properties of the run and cannot be removed after the fact; they and the rest of the archive's measured biases are stated in full in CHAPTER\_KNOWN\_PROPERTIES.md.
- **Below the ground the pressure levels are clamped, not extrapolated.** Where a level

lies under the terrain, the archive repeats the **lowest model level's value verbatim** — matched to exactly 0.0 for  $t$ ,  $1e-9$  for  $q$ . That is a real model state from about 24 to 27 m above the ground rather than synthetic air, and it keeps the surface join coherent. But **\*\*ERA5 extrapolates there with a lapse rate\*\***, so the two archives differ in convention exactly where 1000 hPa is underground, which is most of the land; and our  $z$  at 1000 hPa then sits a median of 24 to 27 m *above* the terrain that  $sp$  puts it below, on **47.6 % of the domain** at 1000 hPa and 9 % at 925. Mask the levels against  $sp$  if you need a clean comparison — every field needed to do it is in the same file.

- **sst** is constant within each 24 h run ( $SST\_UPDATE=0$ ), re-initialised daily.
- **dl** is the lake depth database the run was built with, not a model state: lake physics was off ( $sf\_lake\_physics=0$ ), so nothing in the simulation responds to it. It is published as the boundary dataset it is, and masked off-lake — the raw field is the 10 m WPS default fill on 99.56 % of the domain, and 79 464 of the 88 785 lake cells carry that same default (ERA5's own **dl** does the same where the database is silent).

## 5. Masked fields

Written with a GRIB bitmap rather than fake zeros, so missing means missing:

| field           | defined where                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sst, ci         | <b>sea</b> points only — and sea here is not simply "not land": WPS's land-use class calls the Black Sea, the Sea of Azov and part of the Baltic a <i>lake</i> , so masking on that class would have deleted sst from the Black Sea. The discriminator is connected-component size, measured: the water bodies of this domain are 844 431 cells (Atlantic + Mediterranean + Baltic + North Sea), 49 841 (Black Sea + Azov), 8 583 (Red Sea), then 1 225 (Vänern, the largest genuine lake) and smaller, so the cut sits in a gap with a factor of seven of slack |
| cl              | zero on the sea rather than missing, since a sea point genuinely has no lake cover                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| mucin           | the columns where the CAPE routine returns a value at all, i.e. where CAPE reaches 100 J/kg. <b>That is 4 % to 29 % of the domain</b> , least in winter. The gaps are written as missing rather than zero, because zero would read as "nothing inhibits convection" at precisely the columns where the most does                                                                                                                                                                                                                                                 |
| tsn             | cells whose snow cover exceeds 0.9 ( $\approx$ 29 mm water equivalent); out of season the field is legitimately empty                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| fal             | daytime only (incoming shortwave above 50 W/m <sup>2</sup> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| swvl1-4, stl1-4 | the <b>permanent</b> land of the static file — a fixed outline, the same in every file of the archive. Sea ice therefore carries no soil column, which is what ERA5 does too                                                                                                                                                                                                                                                                                                                                                                                     |
| slt             | land points of the static file (1 304 109 cells)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| field           | defined where                                                                                                                                                                                                                                                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tv <sub>l</sub> | everywhere except the cells where a shrubland dominates the low vegetation (124 672 cells, 15.8 % of those with cv <sub>l</sub> > 0). MODIS carries no shrub phenology while ECMWF's table splits evergreen from deciduous shrubs, and rather than invent one the field is left missing. cv <sub>l</sub> still gives the cover there |
| d <sub>l</sub>  | lake cells only (88 785)                                                                                                                                                                                                                                                                                                             |

## 6. What is not in the archive

No TKE and no boundary-layer height (the PBL scheme is YSU, non-local); no ocean waves, currents, sea surface height or sea-ice thickness (no ocean or wave coupling); no convective precipitation (CU\_PHYSICS=0, so it is identically zero and would be a dead field). Deliberately dropped rather than approximated:

- **leaf area index** (lai<sub>lv</sub>, lai<sub>hv</sub>): the model carries a single LAI per cell, while ERA5 wants one per vegetation class, defined per unit of vegetated area. Any split between the two would be our assumption, so neither field is written. **\*\*The single field itself is in the model output\*\*** — alive, and unpublished for that reason alone; it can be delivered under its own name, LAI, if it is wanted. Note rd<sub>lai2d</sub> = .false., so it is a table lookup on the dominant category rather than an observation, which is a second reason not to split it;
- **sub-grid orography angle and anisotropy** (anor, isor): the static file carries the Kim-Arakawa asymmetry and effective-length parameters, which are not ECMWF's, and the difference does not fit in one line.

Four entries that stood here until 2026-09-17 have moved into the archive, because geo\_em.d02 came back and because the soil objection had an answer:

- **swvl1-4 / stl1-4** are now the true ERA5 layer averages. The land-surface scheme is a *level* scheme — point values at 0, 5, 20, 40, 160 and 300 cm with a linear profile between them — and the whole ERA5 column lies inside those nodes (289 cm is above the deepest, 0 cm is the first). So each layer average is the exact integral of the model's own profile and **nothing is extrapolated**. The objection was against writing the point values, not against integrating them;
- **cv<sub>l</sub> / cv<sub>h</sub>** are sums of the fractional cover over the MODIS-IGBP classes that are low and high vegetation — the same quantity ERA5 means. This is what the dominant category could not express: it holds only 90.1 % of a land cell on average, and 17.9 % of land cells carry both high and low vegetation above 5 %;
- **tv<sub>l</sub> / tv<sub>h</sub>** are the dominant low and high class translated to ECMWF code table 4.234 by the table in §8, where each row is a semantic identity. The one place MODIS cannot answer the question — evergreen versus deciduous shrubs — is left missing rather than guessed;
- **s<sub>lt</sub>** is computed with the FAO clay/sand thresholds that *define* ECMWF's seven soil types, applied to the static file's own clay and sand fractions. No category crosswalk is involved, so the quantity is ERA5's by construction. One caveat for a strict WMO reader: eccodes resolves s<sub>lt</sub> to discipline 2/3/0, which WMO table 4.213 reads as a different 11-value list. This archive is ECMWF semantics throughout, and so is this message.

c<sub>l</sub> and d<sub>l</sub> came with them. c<sub>l</sub> is not redundant with the land-sea mask: 19 309 cells that the mask calls land carry a sub-grid lake, and the model itself discarded the class at run-time. The full reasoning, and the routes by which some of it could still be obtained, is in MISSING\_VARIABLES.pdf.

## 7. The variables

Descriptions and units are the ECMWF ones. "Time" says how the field is sampled. There are 90 rows and 89 distinct shortNames, because z is written twice: as geopotential on the pressure levels and as the surface geopotential (orography). variables=90 messages=246

### Pressure levels (13) — 13 variables, 169 messages

| shortName | paramId | Description                         | Units                          | Level     | Time          |
|-----------|---------|-------------------------------------|--------------------------------|-----------|---------------|
| cc        | 248     | Fraction of cloud cover             | (0-1)                          | 13 levels | instantaneous |
| ciwc      | 247     | Specific cloud ice water content    | kg/kg                          | 13 levels | instantaneous |
| clwc      | 246     | Specific cloud liquid water content | kg/kg                          | 13 levels | instantaneous |
| crwc      | 75      | Specific rain water content         | kg/kg                          | 13 levels | instantaneous |
| cswc      | 76      | Specific snow water content         | kg/kg                          | 13 levels | instantaneous |
| q         | 133     | Specific humidity                   | kg/kg                          | 13 levels | instantaneous |
| r         | 157     | Relative humidity                   | %                              | 13 levels | instantaneous |
| t         | 130     | Temperature                         | K                              | 13 levels | instantaneous |
| u         | 131     | U component of wind                 | m/s                            | 13 levels | instantaneous |
| v         | 132     | V component of wind                 | m/s                            | 13 levels | instantaneous |
| w         | 135     | Vertical velocity (pressure)        | Pa/s                           | 13 levels | instantaneous |
| wz        | 260238  | Geometric vertical velocity         | m/s                            | 13 levels | instantaneous |
| z         | 129     | Geopotential                        | m <sup>2</sup> /s <sup>2</sup> | 13 levels | instantaneous |

### Height above ground — 11 variables, 11 messages

| shortName | paramId | Description                                 | Units | Level | Time                       |
|-----------|---------|---------------------------------------------|-------|-------|----------------------------|
| 100u      | 228246  | 100m U component of wind                    | m/s   | 100 m | instantaneous              |
| 100v      | 228247  | 100m V component of wind                    | m/s   | 100 m | instantaneous              |
| 10fg      | 49      | 10m wind gust (hourly max of resolved wind) | m/s   | 10 m  | max over the previous hour |

| shortName | paramId | Description                          | Units           | Level | Time          |
|-----------|---------|--------------------------------------|-----------------|-------|---------------|
| 10u       | 165     | 10m U component of wind              | m/s             | 10 m  | instantaneous |
| 10v       | 166     | 10m V component of wind              | m/s             | 10 m  | instantaneous |
| 200u      | 228239  | 200m U component of wind             | m/s             | 200 m | instantaneous |
| 200v      | 228240  | 200m V component of wind             | m/s             | 200 m | instantaneous |
| 2d        | 168     | 2m dewpoint temperature              | K               | 2 m   | instantaneous |
| 2r        | 260242  | 2m relative humidity                 | %               | 2 m   | instantaneous |
| 2t        | 167     | 2m temperature                       | K               | 2 m   | instantaneous |
| vwsh      | 260068  | Vertical speed shear (bulk 10-100 m) | s <sup>-1</sup> | 100 m | instantaneous |

#### Mean sea level — 1 variables, 1 messages

| shortName | paramId | Description             | Units | Level          | Time          |
|-----------|---------|-------------------------|-------|----------------|---------------|
| msl       | 151     | Mean sea level pressure | Pa    | mean sea level | instantaneous |

#### Total column — 10 variables, 10 messages

| shortName | paramId | Description                     | Units             | Level  | Time          |
|-----------|---------|---------------------------------|-------------------|--------|---------------|
| hcc       | 188     | High cloud cover                | (0-1)             | column | instantaneous |
| lcc       | 186     | Low cloud cover                 | (0-1)             | column | instantaneous |
| mcc       | 187     | Medium cloud cover              | (0-1)             | column | instantaneous |
| tcc       | 164     | Total cloud cover               | (0-1)             | column | instantaneous |
| tciw      | 79      | Total column cloud ice water    | kg/m <sup>2</sup> | column | instantaneous |
| tclw      | 78      | Total column cloud liquid water | kg/m <sup>2</sup> | column | instantaneous |
| tcrw      | 228089  | Total column rain water         | kg/m <sup>2</sup> | column | instantaneous |
| tcsnw     | 228090  | Total column snow water         | kg/m <sup>2</sup> | column | instantaneous |
| tcw       | 136     | Total column water              | kg/m <sup>2</sup> | column | instantaneous |

| shortName | paramId | Description               | Units             | Level  | Time          |
|-----------|---------|---------------------------|-------------------|--------|---------------|
| tcwv      | 137     | Total column water vapour | kg/m <sup>2</sup> | column | instantaneous |

### Surface / single level — 47 variables, 47 messages

| shortName | paramId | Description                                      | Units             | Level   | Time                     |
|-----------|---------|--------------------------------------------------|-------------------|---------|--------------------------|
| al        | 174     | Albedo (climato-logical, snow-free background)   | (0-1)             | surface | instantaneous            |
| ci        | 31      | Sea ice area fraction                            | (0-1)             | surface | instantaneous            |
| cl        | 26      | Lake cover                                       | (0-1)             | surface | instantaneous, static    |
| cvh       | 28      | High vegetation cover                            | (0-1)             | surface | instantaneous, static    |
| cvl       | 27      | Low vegetation cover                             | (0-1)             | surface | instantaneous, static    |
| fal       | 243     | Forecast albedo (up-ward/downward SW at surface) | (0-1)             | surface | instantaneous            |
| fsr       | 244     | Forecast surface roughness                       | m                 | surface | instantaneous            |
| iews      | 229     | Instantaneous eastward turbulent surface stress  | N/m <sup>2</sup>  | surface | instantaneous            |
| inss      | 230     | Instantaneous northward turbulent surface stress | N/m <sup>2</sup>  | surface | instantaneous            |
| lsm       | 172     | Land-sea mask                                    | (0-1)             | surface | instantaneous            |
| ro        | 205     | Runoff (surface + sub-surface)                   | m                 | surface | accumulated since 00 UTC |
| rsn       | 33      | Snow density                                     | kg/m <sup>3</sup> | surface | instantaneous            |
| sd        | 141     | Snow depth (water equivalent)                    | m                 | surface | instantaneous            |
| sdor      | 160     | Standard deviation of orography                  | m                 | surface | instantaneous            |
| sf        | 144     | Snowfall (water equivalent)                      | m                 | surface | accumulated since 00 UTC |

| shortName | paramId | Description                                      | Units            | Level   | Time                     |
|-----------|---------|--------------------------------------------------|------------------|---------|--------------------------|
| skt       | 235     | Skin temperature (inverted from upward longwave) | K                | surface | instantaneous            |
| slor      | 163     | Slope of orography (resolved, 3 km)              | Numeric          | surface | instantaneous            |
| slt       | 43      | Soil type (ECMWF 7 classes)                      | ~                | surface | instantaneous, static    |
| snowc     | 260038  | Snow cover                                       | %                | surface | instantaneous            |
| sp        | 134     | Surface pressure                                 | Pa               | surface | instantaneous            |
| src       | 198     | Skin reservoir content                           | m                | surface | instantaneous            |
| sro       | 8       | Surface runoff                                   | m                | surface | accumulated since 00 UTC |
| ssr       | 176     | Surface net solar radiation                      | J/m <sup>2</sup> | surface | accumulated since 00 UTC |
| ssrc      | 210     | Surface net solar radiation, clear sky           | J/m <sup>2</sup> | surface | accumulated since 00 UTC |
| ssrd      | 169     | Surface solar radiation downwards                | J/m <sup>2</sup> | surface | accumulated since 00 UTC |
| ssrdc     | 228129  | Surface solar radiation downwards, clear sky     | J/m <sup>2</sup> | surface | accumulated since 00 UTC |
| ssro      | 9       | Sub-surface runoff                               | m                | surface | accumulated since 00 UTC |
| sst       | 34      | Sea surface temperature                          | K                | surface | instantaneous            |
| stl1      | 139     | Soil temperature level 1 (0-7 cm)                | K                | surface | instantaneous            |
| stl2      | 170     | Soil temperature level 2 (7-28 cm)               | K                | surface | instantaneous            |
| stl3      | 183     | Soil temperature level 3 (28-100 cm)             | K                | surface | instantaneous            |
| stl4      | 236     | Soil temperature level 4 (100-289 cm)            | K                | surface | instantaneous            |

| shortName | paramId | Description                                      | Units                          | Level   | Time                     |
|-----------|---------|--------------------------------------------------|--------------------------------|---------|--------------------------|
| str       | 177     | Surface net thermal radiation                    | J/m <sup>2</sup>               | surface | accumulated since 00 UTC |
| strc      | 211     | Surface net thermal radiation, clear sky         | J/m <sup>2</sup>               | surface | accumulated since 00 UTC |
| strd      | 175     | Surface thermal radiation downwards              | J/m <sup>2</sup>               | surface | accumulated since 00 UTC |
| strdc     | 228130  | Surface thermal radiation downwards, clear sky   | J/m <sup>2</sup>               | surface | accumulated since 00 UTC |
| swvl1     | 39      | Volumetric soil water layer 1 (0-7 cm)           | m <sup>3</sup> /m <sup>3</sup> | surface | instantaneous            |
| swvl2     | 40      | Volumetric soil water layer 2 (7-28 cm)          | m <sup>3</sup> /m <sup>3</sup> | surface | instantaneous            |
| swvl3     | 41      | Volumetric soil water layer 3 (28-100 cm)        | m <sup>3</sup> /m <sup>3</sup> | surface | instantaneous            |
| swvl4     | 42      | Volumetric soil water layer 4 (100-289 cm)       | m <sup>3</sup> /m <sup>3</sup> | surface | instantaneous            |
| tirf      | 235015  | Time integral of rain flux (rain only)           | kg/m <sup>2</sup>              | surface | accumulated since 00 UTC |
| tp        | 228     | Total precipitation                              | m                              | surface | accumulated since 00 UTC |
| tsn       | 238     | Temperature of snow layer                        | K                              | surface | instantaneous            |
| tvh       | 30      | Type of high vegetation (ECMWF code table 4.234) | ~                              | surface | instantaneous, static    |
| tvL       | 29      | Type of low vegetation (ECMWF code table 4.234)  | ~                              | surface | instantaneous, static    |
| z         | 129     | Geopotential (surface)                           | m <sup>2</sup> /s <sup>2</sup> | surface | instantaneous            |
| zust      | 228003  | Friction velocity                                | m/s                            | surface | instantaneous            |

“static” in the Time column means the field is identical in every file of the archive: it comes from the run’s static input, not from the hourly state. `lsm` is deliberately NOT static — the model reclassifies sea points as land where sea ice appears, so it changes with the season.

### Entire lake — 1 variable, 1 message

| shortName | paramId | Description      | Units | Level      | Time                  |
|-----------|---------|------------------|-------|------------|-----------------------|
| d1        | 228007  | Lake total depth | m     | entireLake | instantaneous, static |

### Top of atmosphere / most unstable parcel — 7 variables, 7 messages

| shortName | paramId | Description                          | Units            | Level                | Time                     |
|-----------|---------|--------------------------------------|------------------|----------------------|--------------------------|
| mucap     | 228235  | Most-unstable CAPE                   | J/kg             | most unstable parcel | instantaneous            |
| mucin     | 228236  | Most-unstable CIN                    | J/kg             | most unstable parcel | instantaneous            |
| tisr      | 212     | TOA incident solar radiation         | J/m <sup>2</sup> | top of atmosphere    | accumulated since 00 UTC |
| tsr       | 178     | Top net solar radiation              | J/m <sup>2</sup> | top of atmosphere    | accumulated since 00 UTC |
| tsrc      | 208     | Top net solar radiation, clear sky   | J/m <sup>2</sup> | top of atmosphere    | accumulated since 00 UTC |
| ttr       | 179     | Top net thermal radiation            | J/m <sup>2</sup> | top of atmosphere    | accumulated since 00 UTC |
| ttrc      | 209     | Top net thermal radiation, clear sky | J/m <sup>2</sup> | top of atmosphere    | accumulated since 00 UTC |

## 8. How the land-surface categories were translated

tv1, tvh and slt are the only fields in the archive that carry a *code* rather than a number, so the tables that produced them are published here: without them the codes cannot be read, and the mapping cannot be checked. The source is the run's own static file geo\_em.d02 (MODIS-IGBP 21 classes, WPS MMINLU=MODIFIED\_IGBP\_MODIS\_NOAH). Its land use was verified against the hourly output: it agrees with the model's dominant category on **100.00 %** of points once one expected difference is excluded — the model recodes lake to water at runtime, because lake physics was off. This is what makes cl recoverable at all.

### tv1 / tvh: MODIS-IGBP → ECMWF code table 4.234

High and low follow the IGBP class definitions of tree cover, not a canopy-height table.

| MODIS-IGBP class              | set  | ECMWF 4.234                                         |
|-------------------------------|------|-----------------------------------------------------|
| 1 Evergreen Needleleaf Forest | high | 3 Evergreen needleleaf trees                        |
| 2 Evergreen Broadleaf Forest  | high | 6 Evergreen broadleaf trees                         |
| 3 Deciduous Needleleaf Forest | high | 4 Deciduous needleleaf trees                        |
| 4 Deciduous Broadleaf Forest  | high | 5 Deciduous broadleaf trees                         |
| 5 Mixed Forests               | high | 18 Mixed forest/woodland                            |
| 8 Woody Savannas              | high | 19 Interrupted forest (both are 30-60 % tree cover) |
| 9 Savannas                    | low  | 7 Tall grass                                        |

| <b>MODIS-IGBP class</b>                                 | <b>set</b> | <b>ECMWF 4.234</b>                                          |
|---------------------------------------------------------|------------|-------------------------------------------------------------|
| 10 Grasslands                                           | low        | 2 Short grass                                               |
| 11 Permanent Wetlands                                   | low        | 13 Bogs and marshes (absent from this domain)               |
| 12 Croplands                                            | low        | 1 Crops, mixed farming                                      |
| 14 Cropland/Natural Mosaic                              | low        | 1 Crops, mixed farming (absent from this domain)            |
| 18 Wooded Tundra, 19 Mixed Tundra, 20 Barren Tundra     | low        | 9 Tundra                                                    |
| <b>6 Closed Shrublands, 7 Open Shrublands</b>           | low        | <b>not translated — tvl is missing there</b>                |
| 13 Urban, 15 Snow and Ice, 16 Barren, 17 Water, 21 Lake | neither    | — (not vegetation; they are the remainder of 1 - cvl - cvh) |

Every translated row is a semantic identity: the ECMWF type names the same vegetation as the MODIS class. The two shrubland classes are the exception and are the reason tvl carries a mask. MODIS does not record whether a shrubland is evergreen or deciduous, while table 4.234 has separate codes for the two (16 and 17). Choosing one would have been our invention over 16.2 % of the vegetated cells, so the field is left missing there instead; cvl still reports the cover. The classes that would have been the other ambiguous rows, 11 and 14, do not occur in this domain at all.

#### **slt: computed, not translated**

No category crosswalk is involved. ECMWF's seven soil types are *defined* by clay and sand percentage thresholds, and the static file carries clay and sand fractions, so the definition is applied directly:

| <b>condition (percent)</b>                                                    | <b>ECMWF soil type</b> |
|-------------------------------------------------------------------------------|------------------------|
| clay $\geq$ 60                                                                | 5 very fine            |
| $35 \leq$ clay < 60                                                           | 4 fine                 |
| clay < 35 and sand $\leq$ 15                                                  | 3 medium fine          |
| ( $18 \leq$ clay < 35 and sand > 15) or (clay < 18 and $15 <$ sand $\leq$ 65) | 2 medium               |
| clay < 18 and sand > 65                                                       | 1 coarse               |
| dominant soil category is <i>organic material</i>                             | 6 organic              |

Every land point classifies; type 7 (tropical organic) does not occur here. Distribution over the 1 304 109 land points: 1 069 456 medium, 187 378 coarse, 45 450 fine, 1 498 organic, 327 very fine.

#### **swvl1-4 / stl1-4: integrated, not relabelled**

The land-surface scheme is a *level* scheme: it carries point values at 0, 5, 20, 40, 160 and 300 cm, with a linear profile between them. ERA5 wants averages over 0–7, 7–28, 28–100 and 100–289 cm. Those four intervals lie entirely inside the model's own nodes — 289 cm is above the deepest one and 0 cm is the first — so each ERA5 layer is the **exact integral** of the model's profile over it, and nothing is extrapolated at either end. In practice this is a fixed 4 × 6 matrix of weights, applied to the six model levels. swvl uses the total soil moisture (liquid plus ice), which is what the ERA5 name means; the liquid-only field is not used. Both sets are masked over water.